

Shared Priors, Not Assistance in General: What Public Co-Writing Data Support about Homogenization

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Abstract

When many people write with the same language model, two risks are often joined into one story: outputs become more alike, and later unaided skill weakens. We separate that package and test what public microdata can support. Re-analyzing four released corpora, we find a narrow positive result and several negative or non-identified ones. In the Padmakumar and He co-writing essays, InstructGPT reduces within-topic diversity relative to solo writing and relative to GPT-3; model spans under InstructGPT are less diverse than user spans. Topic-level permutation tests reject the null for both InstructGPT contrasts ($p = 0.003$ and $p = 0.001$), while the solo-versus-GPT-3 contrast does not ($p = 0.71$). Vectorizer robustness checks preserve the same ordering. CoAuthor shows no strong acceptance dose-response on group diversity. Moon et al. materials show a concentrated model prior, but not a clean post-ChatGPT lexical collapse. Public study-companion logs cannot identify transfer debt. We keep prior-driven homogenization, drop assistance-as-such claims, and leave debt open.

CCS Concepts: Human-centered computing → Empirical studies in HCI.

Keywords: large language models, homogenization, shared priors, co-writing, open reanalysis

1 Introduction

Language models improve local drafting. They also create a population question. If many writers condition on one generative prior, the set of answers across people can shrink even while each person feels faster [Padmakumar and He, 2024, Anderson et al., 2024, Acemoglu et al., 2026]. A second question is cognitive debt: weaker performance once the helper is removed [Kosmyna et al., 2025, Barcaui, 2025, Risko and Gilbert, 2016, Parasuraman and Riley, 1997].

These worries are often told as one mechanism, cognitive convergence: offloading onto a shared prior that both compresses diversity and leaves a transfer cost. That framing is a research program, not a finding, until public data can carry it.

This paper is a falsification-oriented reassessment of four public corpora. We rank claims by support:

- Keep: shared feedback-tuned priors can compress co-written diversity.
- Drop: any LM assistance homogenizes.
- Leave open: transfer debt as part of the same mechanism.

Contributions. (1) A controlled reanalysis of Padmakumar and He [Padmakumar and He, 2024] with topic bootstrap intervals, permutation tests, and vectorizer robustness. (2) A CoAuthor dose check that does not confirm strong acceptance-driven group homogenization [Lee et al., 2022]. (3) A Moon et al. contrast between a concentrated model prior and mixed observational era effects [Moon et al., 2026a,b]. (4) An explicit non-identification result for debt in public study-companion logs [Simsek et al., 2025, Anders et al., 2025].

2 Related Work

2.1 Homogenization under shared generative priors

A growing literature asks whether language-model assistance compresses the set of ideas or texts that people produce. Padmakumar and He [Padmakumar and He, 2024] provide the cleanest public contrast for writing: argumentative essays under solo, GPT-3, and InstructGPT conditions, with sentence-level authorship masks. Their design matters because it separates feedback-tuned assistance from base-model assistance, which is exactly the split our reanalysis exploits. Anderson et al. [Anderson et al., 2024] report homogenization in creative ideation, but without public microdata for independent reanalysis. Moon et al. [Moon et al., 2026a,b] study admissions writing and separate surface (lexical) diversity from conceptual originality. Their released materials include matched human versus model comparisons and within-subject unaided, GPT-revised, and GPT-generated essays, which lets us test whether the model prior itself is narrow even when observational era contrasts are mixed.

At the population scale, Acemoglu et al. [Acemoglu et al., 2026] formalize knowledge collapse when many agents condition on similar AI outputs. That theory motivates our question, but it does not identify which public co-writing corpora already show compression.

2.2 Co-writing systems and interaction dose

Lee et al. [Lee et al., 2022] released CoAuthor, a large GPT-3 co-writing log with suggestion requests and acceptances. The corpus is valuable for studying how people use suggestions, but it has no unaided writing arm and therefore cannot estimate homogenization relative to solo baselines. We use it only for a weaker dose question: whether higher acceptance rates coincide with lower cross-session diversity. Dell’Acqua et al. [Dell’Acqua et al., 2025] show that AI can raise productivity and quality on some knowledge tasks and hurt them on others. That jagged frontier is about task performance, not cross-person diversity; we keep the distinction explicit so performance gains are not mistaken for population variety.

2.3 Offloading, over-reliance, and transfer debt

Cognitive offloading [Risko and Gilbert, 2016] and automation misuse [Parasuraman and Riley, 1997] predate LLMs and warn that helpers can change what people encode and later recall. Kosmyna et al. [Kosmyna et al., 2025] report EEG and behavioral signs of cognitive debt when writers rely on ChatGPT for essays. Barcaui [Barcaui, 2025] finds retention costs in a randomized trial framed as a cognitive crutch. Lee et al. [Lee et al., 2025] survey knowledge workers who report lower perceived effort and confidence shifts under generative AI. These studies support debt as a live hypothesis, but most do not release microdata, and none of our public corpora identify debt versus unaided learning by themselves.

Şimşek et al. [Şimşek et al., 2025, Anders et al., 2025] do release study-companion logs with retention and transfer scores. Both processed arms still use ChatGPT, so the public files cannot answer whether AI assistance creates transfer debt relative to unaided study. We treat that as a non-identification result, not a null finding.

2.4 What this paper adds

Prior work often joins homogenization and debt into one story. We separate them and ask what four public corpora can support. Relative to Padmakumar and He [Padmakumar and He, 2024], we add topic-level bootstrap intervals, permutation tests, and vectorizer robustness. Relative to CoAuthor [Lee et al., 2022] and Moon et al. [Moon et al., 2026a], we report negative or mixed checks that bound overclaims. Relative to debt studies [Kosmyna et al., 2025, Barcaui, 2025, Şimşek et al., 2025], we state when public logs cannot identify the claim at all.

3 Claims

Hypothesis 1 (H1). Within-topic diversity is lower for InstructGPT essays than for solo essays.

Hypothesis 2 (H2). Within-topic diversity is lower for InstructGPT than for GPT-3.

Hypothesis 3 (H3). Under InstructGPT, model spans are less diverse than user spans.

RQ 1 (RQ1). In CoAuthor, does higher suggestion acceptance reduce cross-session diversity?

RQ 2 (RQ2). In Moon et al. materials, is the model prior narrower than unaided human writing, and do observational post-ChatGPT essays show lower diversity?

RQ 3 (RQ3). Can public study-companion logs identify transfer debt versus unaided learning?

4 Method

4.1 Data

Padmakumar and He [Padmakumar and He, 2024]: after dropping essays under 40 words, solo $N = 99$, GPT-3 $N = 97$, InstructGPT $N = 100$, across ten topics. User and model texts use authorship labels U and A.

CoAuthor [Lee et al., 2022]: final documents with at least 40 words ($N = 1218$); acceptance rate among sessions with at least one suggestion request ($N = 1185$).

Moon et al. [Moon et al., 2026a,b]: released pre/post aggregates, matched within-prompt distances, and within-subject unaided / GPT-revised / GPT-generated essays ($N = 203$ per condition).

Şimşek et al. [Şimşek et al., 2025, Anders et al., 2025]: processed scores for summary versus transcript ($N = 101$) and summary versus question ($N = 215$).

4.2 Measures

Primary diversity for Padmakumar and CoAuthor is one minus mean pairwise TF-IDF cosine similarity within a stratum (topic or acceptance tercile), with a vectorizer fit on the relevant

corpus. Padmakumar also reports MiniLM embedding diversity. Homogenization versus solo is $H = (D_{\text{solo}} - D)/D_{\text{solo}}$. For Moon materials we use released word and document distances. For learning logs we compare transfer by condition and help-count correlations within arm.

4.3 Inference for Study 1

We treat topics as the resampling unit. For each condition we compute within-topic diversity, then draw 5000 topic bootstraps to form 95% intervals for mean D and for pairwise gaps. We also run one-sided essay-label permutation tests within topics (999 permutations): under the null, solo and InstructGPT essays are exchangeable within topic. Robustness checks recompute the three condition means with unigram TF-IDF and with stop-words retained. Analysis code: `scripts/study1_stats.py`.

5 Results

5.1 Study 1: the claim that holds

Within-topic TF-IDF diversity is 0.780 (solo), 0.788 (GPT-3), and 0.740 (InstructGPT), so $H = 0.051$ for InstructGPT versus solo (Table 1). Embedding diversity falls from 0.359 to 0.329 (GPT-3) and 0.298 (InstructGPT).

Table 1: Study 1 main estimates (Padmakumar and He).

Condition	N	TF-IDF D	Emb. D	H
solo	99	0.780	0.359	0.000
gpt3	97	0.788	0.329	-0.010
instructgpt	100	0.740	0.298	0.051

Table 2 reports topic-bootstrap gaps. The GPT-3 minus InstructGPT gap is 0.047 (95% CI [0.001, 0.097]; permutation $p = 0.001$). The solo minus InstructGPT gap is 0.040 (CI [-0.018, 0.099]; $p = 0.003$); 90.7% of bootstrap replicates are positive. The solo minus GPT-3 gap is near zero ($p = 0.71$). H1 and H2 are supported by the permutation tests and by the GPT-3 contrast interval; the solo-InstructGPT interval is wide because only ten topics are available.

Table 2: Study 1 diversity gaps (topic bootstrap 95% CIs; one-sided permutation p).

Contrast	Est.	CI lo	CI hi	p
solo - IGPT	0.040	-0.018	0.099	0.003
GPT-3 - IGPT	0.047	0.001	0.097	0.001
solo - GPT-3	-0.008	-0.063	0.046	0.709

Under InstructGPT, model spans are less diverse than user spans (Table 3: TF-IDF 0.820 vs 0.832; embeddings 0.316 vs 0.364). Under GPT-3, model spans are more diverse than user spans. H3 is directional; the InstructGPT authorship gap is small and its topic-bootstrap interval includes zero.

Table 3: Authorship split (within-topic diversity).

Condition	N	TF-IDF D	Emb. D
gpt3 user	96	0.849	0.361
gpt3 model	94	0.897	0.453
instructgpt user	96	0.832	0.364
instructgpt model	98	0.820	0.316

Robustness. Unigram TF-IDF and stop-word retention preserve the same order: InstructGPT < solo $\approx$ GPT-3 (Table 4).

Table 4: Vectorizer robustness (mean within-topic D).

Spec	solo	gpt3	instructgpt
main (1–2 gram)	0.780	0.788	0.740
unigram	0.743	0.748	0.700
no stop-word filter	0.726	0.730	0.692

5.2 Study 2: the dose pattern that does not

In CoAuthor, group diversity is almost flat across acceptance terciles: 0.929, 0.929, and 0.922 (Table 5). Acceptance correlates only mildly with similarity to the corpus centroid ($\rho = +0.178$). Never-request sessions are rare ($N = 33$) and are not a clean solo control. RQ1: weak atypicality signal, no strong group-diversity dose curve.

Table 5: CoAuthor by acceptance tercile.

Bin	N	Accept rate	TF-IDF D
low	395	0.427	0.929
mid	395	0.731	0.929
high	395	0.930	0.922

5.3 Study 3: a narrow model prior, mixed era effects

Matched within-prompt document distance is 0.302 for human essays versus 0.116 for GPT essays. Within subjects, document distance falls from 0.293 (unaided) to 0.219 (GPT-revised) and 0.178 (GPT-generated); 92%–95% of people are lower under GPT than under their own unaided essay (Table 6).

Table 6: Moon et al. distance summaries.

Panel	Condition	N	Doc. distance
Matched	human	n/a	0.302
Matched	gpt	n/a	0.116
Within	unaided	203	0.293
Within	gpt revised	203	0.219
Within	gpt generated	203	0.178

Observational era contrasts are mixed. Word distance is slightly higher after ChatGPT (0.832 vs 0.830), while document distance is slightly lower (0.214 vs 0.222). These are calendar labels, not individual AI-use labels. RQ2: model prior narrowing is supported; population lexical collapse is not.

5.4 Study 4: debt is not identified here

Both arms in each Şimşek et al. study use ChatGPT [Şimşek et al., 2025]. Transfer means differ by roughly 0.1–0.2 points across arms; help-count correlations with transfer are near zero. RQ3: not testable as debt versus unaided learning. Debt estimates in Kosmyna et al. [Kosmyna et al., 2025] and Barcaui [Barcaui, 2025] remain external.

5.5 Claim map

Table 7 and Figure 1 summarize the four-corpus evidence.

Table 7: What public microdata support.

Claim	Verdict	Where
InstructGPT homogenizes vs solo/GPT-3	Supported	Padmakumar
Any LM assistance homogenizes	Not supported	Padmakumar
Accept dose on group diversity	Weak / mixed	CoAuthor
Model-only text narrower than human	Supported	Moon et al.
Post-ChatGPT word-diversity collapse	Not supported	Moon et al.
Transfer debt in public learning logs	Not testable	Şimşek et al.
Unified cognitive convergence	Not established	All

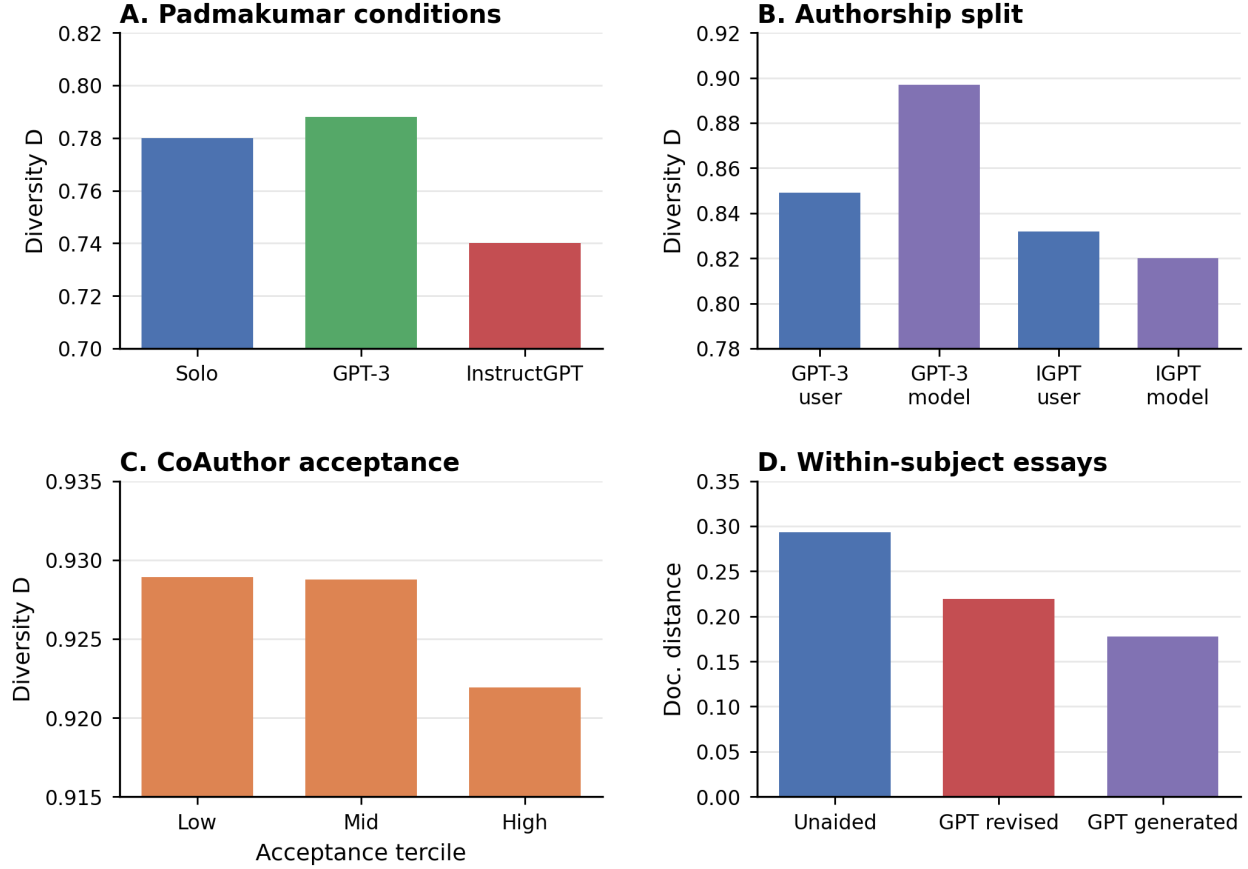


Figure 1: Evidence map across corpora. (A) InstructGPT lowers diversity; GPT-3 does not. (B) Under InstructGPT, model spans are the less diverse component. (C) CoAuthor acceptance barely moves group diversity. (D) GPT-generated and GPT-revised essays are narrower than unaided essays.

6 Discussion

The result worth keeping is precise. A shared feedback-tuned prior can pull co-written essays closer together [Padmakumar and He, 2024]. Permutation tests and the GPT-3 contrast interval back that claim. Base GPT-3 does not reproduce the drop, so homogenization is better explained by the prior people share than by help as such.

The larger cognitive-convergence package does not earn the same status. CoAuthor lacks a clear acceptance-to-diversity dose curve [Lee et al., 2022]. Moon et al. show that the model itself is a narrow attractor, while observational word diversity does not fall after ChatGPT [Moon et al., 2026a]. Text homogenization is not proof of offloading. Debt remains a separate question wherever unaided controls are missing.

For evaluation practice, report three quantities apart: assisted quality, cross-person diversity under a shared tool, and tool-removed transfer.

7 Limitations

Padmakumar spans ten topics, so topic-bootstrap intervals are wide; the solo versus InstructGPT CI includes zero even though the permutation test is significant. Authorship compression under InstructGPT is directional but not tightly estimated. CoAuthor dose analyses are correlational and lack solo writing. Moon observational contrasts confound year with possible AI use. Learning-companion arms have no no-AI control. This paper is a public-data reassessment, not a new human experiment.

8 Conclusion

Public co-writing data support a limited claim: shared feedback-tuned priors can compress writing diversity. They do not support a blanket claim that assistance homogenizes, and they do not establish unified cognitive convergence with proven transfer debt. Keep the prior result. Drop the overclaim. Leave debt to designs that can actually test it.

Data and code

Upstream corpora:

- Padmakumar and He: <https://github.com/vishakhpk/hai-diversity>
- CoAuthor: <https://coauthor.stanford.edu>
- Moon et al.: <https://doi.org/10.17605/OSF.IO/YD94Z>
- Study companion: <https://doi.org/10.17605/OSF.IO/SWYTK>

Analysis code, paper tables, and figure scripts:

<https://github.com/baban9/shared-priors-homogenization>

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Competing interests

The author declares no competing interests.

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